

Unequal Mobility and Residential Segregation: A Schelling-Type Agent-Based Model of Relocation Constraints

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1 Introduction

1.1 Motivation & Related Work

Our research is based on the landmark paper “Dynamic Models of Segregation” (Schelling, 1971) which modelled how segregation occurs in neighborhoods, even when individuals don’t have any strong preferences towards living in a segregated community. He postulated a theory of ‘tipping’, where after a certain number of black people move into a neighborhood, the white people will flee. This was based on previous research, which he paraphrased as the observation that „most Americans can tolerate about 20 percent blacks in their community“ before they move out. As black people begin to fill up remaining occupancies, white people begin to flee.

Schelling’s own research found no neighborhoods which could retain a 50-50 split in residences, but that once a community reached a point of 75% majority to 25% minority, the balance would irreversibly continue shifting until the minority became the majority, as the majority would flee. Even in models with increased disposition towards tolerance (e.g., where whites are okay being a one-third minority), the model always tends towards segregation and provides evidence that segregation is significantly more likely to occur than integration.

We wanted to create a model that takes Schelling’s original model and extends it to account for unequal access to housing options, financial constraints, information barriers, and discrimination or restricted residential mobility. Although the original model provides valuable insight into how people’s preferences shape the diversity of a community, it fails to provide any insight into how people’s opportunities affect their living decisions, other than by lack of available housing. With this being a very influential study, however, there have of course already been researchers who have helped fill in certain gaps in the original research.

In the paper "Combining segregation and integration: Schelling model dynamics for heterogeneous population" (Hatna & Benenson, 2015), the authors demonstrate a mixed-integration effect previously undiscussed in Schelling’s original study. This mixed-integration effect shows what happens when some residents in an area have a high tolerance, and others have a low tolerance. In this case, ‘patches’ begin to emerge where some residents group themselves with others who share their identification, and some group themselves diversely in bundles with others who are dislike themselves. This is mostly because intolerant individuals will leave to group themselves with those who are alike, whereas tolerant individuals much more rarely move, especially if they have a disposition against intolerant individuals.

Building off the general concept of agent-based modeling, the paper "From Factors to Actors: Computational Sociology and Agent-Based Modeling" (Macy & Wiler, 2002) explores and analyzes the impact of agent-based models on sociological research. Although it does not directly add

to the research done by Schelling, it provides a critique about models similar to Schelling's and suggestions for future applications of agent based modelling, such as the need for such models to increase "robustness" and to "avoid reliance on biological metaphors". While the 'tolerance' of an individual towards others is a very real factor in segregation, it is something hard to accurately measure in the real world, especially due to messy bias in self-reported measures (Rosenman et al., 2011). The factors we chose to add to our version of the segregation model are based on purely quantitative, non-self-reported factors affecting individuals, which should ideally generate a model more easily transferable to real world analysis than the assumptions about how people feel about others.

The paper "Causal Mechanisms in the Social Sciences" (Hedström et al., 2010) further elaborates on the importance and potential uses of agent based modelling in sociological research. The authors describe agent-based models as being valuable in exploring two types of facts: empirical and 'stylized'. 'Stylized' facts are those which "do not necessarily have close resemblance to any particular empirical fact" (Hedström et al., 2010), but nonetheless are still important in exploring causes of empirical facts and explanations for phenomena in the real world. In essence, the exploration of 'stylized facts' allows researchers and, in the case of their paper, philosophers, to delve deeper into real problems through computerized thought experiments and lay greater cause to the claims a scientist or philosopher is making, such as in the case of our agent based model, where we wish to show how factors like income inequality impact segregation.

Likewise, the importance of our study is not merely to create a thought experiment through 'stylized facts', but rather to extend it to already assessed empirical studies. Housing can be understood as "the context within which choice and mobility occur, but the choices and the ability to move are conditioned by financial institutions and government policies" (Clark, 2017), and especially of relevance to us is how high quality housing is often of "limited affordability for minority households" (Clark, 2017). Although the limitations of housing mobility for minorities in a socio-economic context has been widely observed, there is a gap in literature evaluating several effects using agent based modelling, which we hope to address in this paper.

1.2 Research Questions

Using our adapted model, we seek to address the following research question:

Does unequal mobility between social groups increase final segregation, convergence time, satisfaction inequality and relocation constraints in a Schelling-type residential model?

With the additional sub-questions to also hopefully be answered:

- **RQ1:** Does lower mobility for one group increase the final residential segregation index?
- **RQ2:** Does lower mobility for one group increase the number of steps needed to reach equilibrium?
- **RQ3:** Does lower mobility for one group increase the satisfaction gap between the high-mobility and low-mobility groups?
- **RQ4:** Does lower mobility for one group increase failed relocation attempts for that group?
- **RQ5:** Does a higher tolerance threshold strengthen the effect of unequal mobility on convergence time and satisfaction inequality?
- **RQ6:** Does restricted destination access, such as radius-limited movement, increase the effect of unequal mobility on convergence time and satisfaction inequality?

We hypothesize that unequal opportunities to move will increase the final segregation, increase the number of steps needed to reach equilibrium, increase the satisfaction gap between the two groups, increase failed relocation attempts by the group with lower mobility, and that the effects of unequal mobility will actually be more strongly demonstrated when agents have higher tolerance thresholds.

2 Methodology

Schelling's original model plots Xes and Os on a grid and evaluates their placement over several 'turns', with each turn being an opportunity for individual agents to move somewhere more to their preference. This was originally based solely on their tolerance for being positioned around other agents which are either similar or dissimilar to themselves, as well as available spaces. For our model, we created a 40x40 grid, with each crosspoint representing a potential agent. Each space can be filled by a single agent, or remain empty. Agents are randomly assigned to a space at the beginning of the simulation, and each turn has the possibility to move. 10% of the spaces are left empty.

Our agents are split into two groups, A and B: Group A has no restrictions on their ability to move (mobility probability = 1), whereas members of group B are restricted in varying degrees (mobility probability can be 1, 0.8, 0.6, 0.4, or 0.2). Furthermore, for RQ6 group B can only move a maximum 6 spaces away. This has been programmed to represent various inequalities with regard to access to finances, housing availability, and/or discrimination. The agents also all have a tolerance threshold, which is measured as a ratio of similar neighbors to dissimilar neighbors. For example, if an agent has a tolerance threshold of 0.3, then they will seek to move if more than 30% of their surrounding neighbors are dissimilar to them. They become increasingly dissatisfied the more their threshold tolerance is broken and become increasingly likely to move. The simulation runs until either all agents are satisfied, or until 300 turns have been processed.

3 Results

3.1 Example simulation

As a first illustrative model, given a group A mobility of 1 and a group B mobility of 0.4 and using a tolerance of 0.5 creates the following pattern as a final grid:

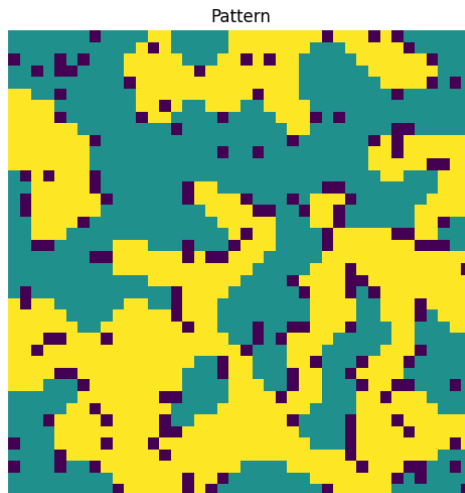


Figure 1: Arrangement of cells after the model converges.

Additionally, we can see relationships between agents over all steps, such as relocations, segregation and satisfaction:

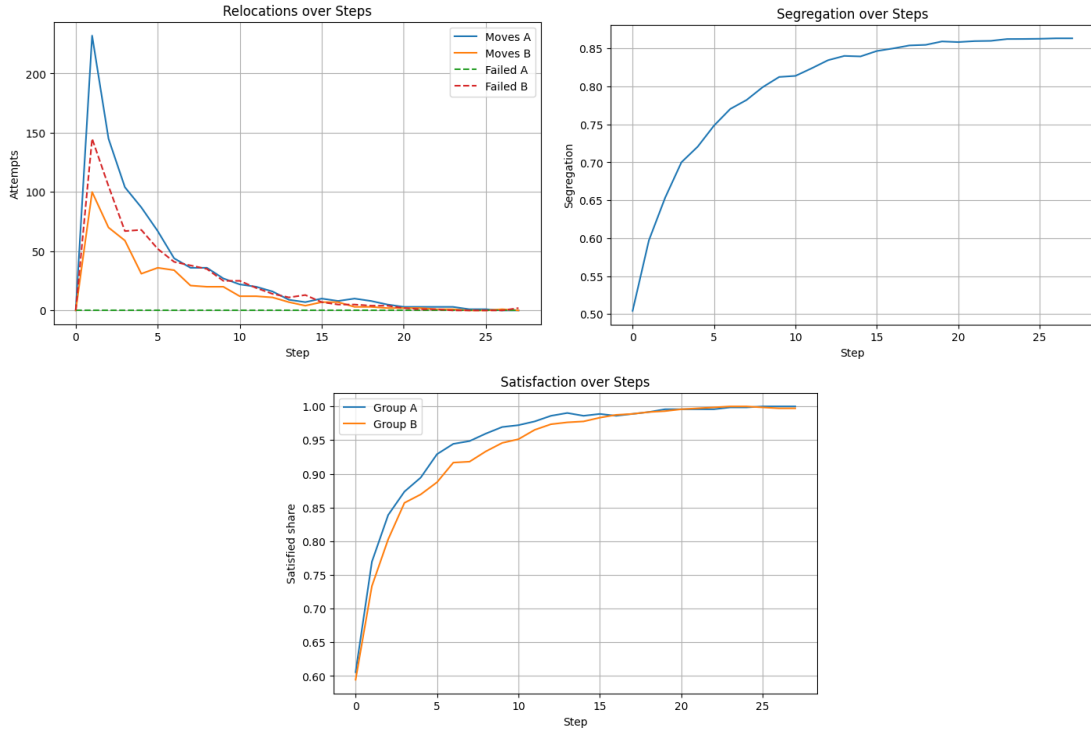


Figure 2: Relocations, Segregation and Satisfaction over time.

As can be seen in these plots, Group B has a higher amount of failed moves, while also having a lower amount of total moves, both of which decline over the amount of steps. Segregation becomes stable at roughly 0.85 after 22 steps. The same happens to satisfaction, with group B having a slightly lower satisfaction in the beginning, but both groups converging to 100% satisfaction after 25 steps.

3.2 Simulations

The simulations were run with the following parameters set, totaling 2000 simulations over all variants.

Table 1: Parameter settings used for the simulation experiments.

Parameter	Value
Grid size	40×40
Density	0.90
Empty cells	10%
Group share	50% Group A, 50% Group B
Neighborhood	Moore neighborhood, 8 surrounding cells
Maximum steps	300
Repetitions	50 per condition
Group A mobility	1.0
Group B mobility	1.0, 0.8, 0.6, 0.4, 0.2
Tolerance threshold	0.3, 0.4, 0.5, 0.6
Destination modes	anywhere, radius($r=6$)

3.3 Segregation through lower mobility(RQ1)

Figure 3 shows the distribution of final segregation over all turns for all possible mobilities in group B.

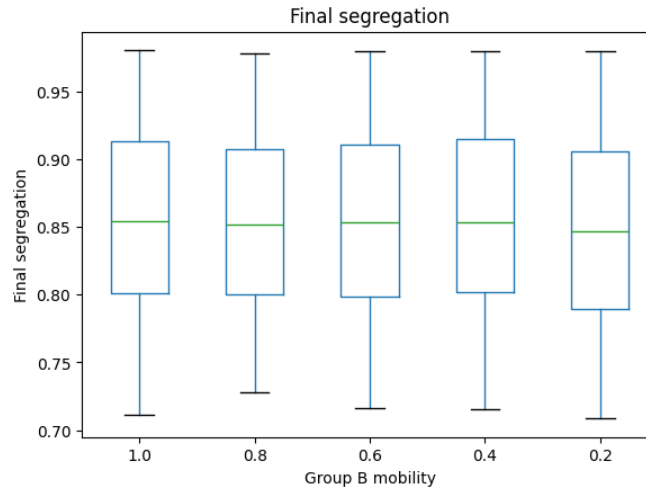


Figure 3: Boxplot of final segregation for all group B mobilities.

As can be seen, for all possible mobility statistics of agents in group B, the final segregation remains very similar, with a slightly lower median for a mobility of 0.2. However, every box shows similar traits, with some boxes having shorter whiskers, but no significant differences. The highest average final segregation is 0.856, with the lowest being 0.852, giving an average final segregation difference of 0.004.

These findings show that RQ1 is not supported in the main model; lower mobility for group B does not meaningfully increase the final segregation.

3.4 Equilibrium through lower mobility(RQ2)

In Figure 4, again a boxplot can be seen, this time showing the amount of steps needed to equilibrium(both groups satisfied and stable or no moves happen) for all simulations.

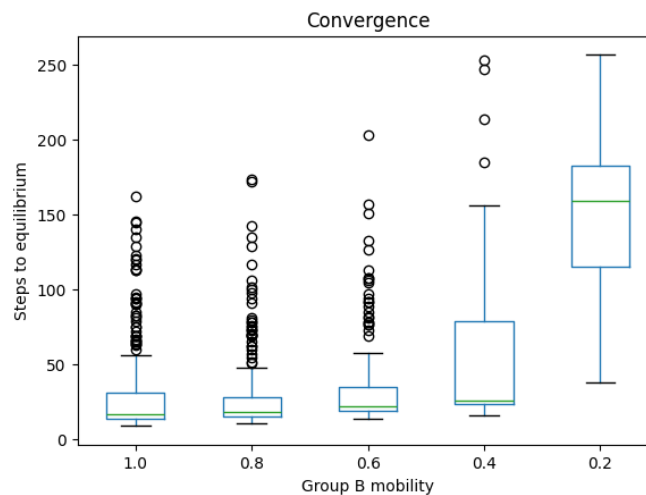


Figure 4: Boxplot of convergence for all group B mobilities.

It can be seen that a higher mobility for group B causes a faster convergence time, with only a few outliers taking over 60 steps for a mobility of 0.6 - 1. At a mobility of 0.4 we start to see an increase in steps to reach convergence, with a mobility of 0.2 increasing steps even further. The lowest average amount of steps to equilibrium was found at 33.11 steps and the highest average amount was found at 149.4 steps.

3.5 Satisfaction through lower mobility(RQ3)

Figure 5 shows the satisfaction gap for different mobilities of group B.

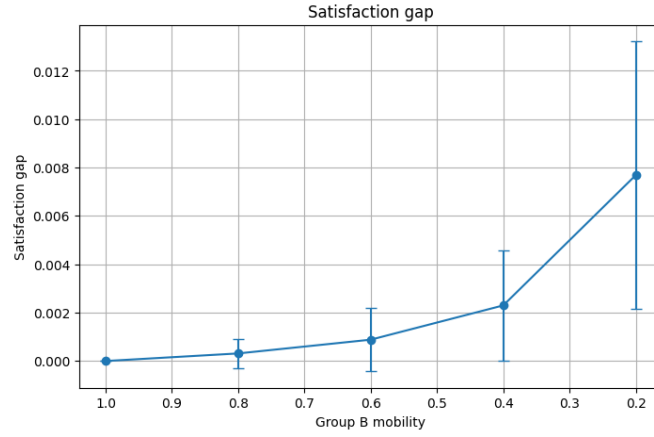


Figure 5: Satisfaction gap for different mobilities. Higher gap means lower satisfaction for group B.

Here we can see a clear increase in satisfaction gap when group B has mobility. A group B with mobility of 1 is the baseline in this case, with each decrease in mobility clearly showing an increase in satisfaction gap.

3.6 Relocation attempts through lower mobility(RQ4)

Figure 6 shows the amount of successful and failed moves for different mobilities.

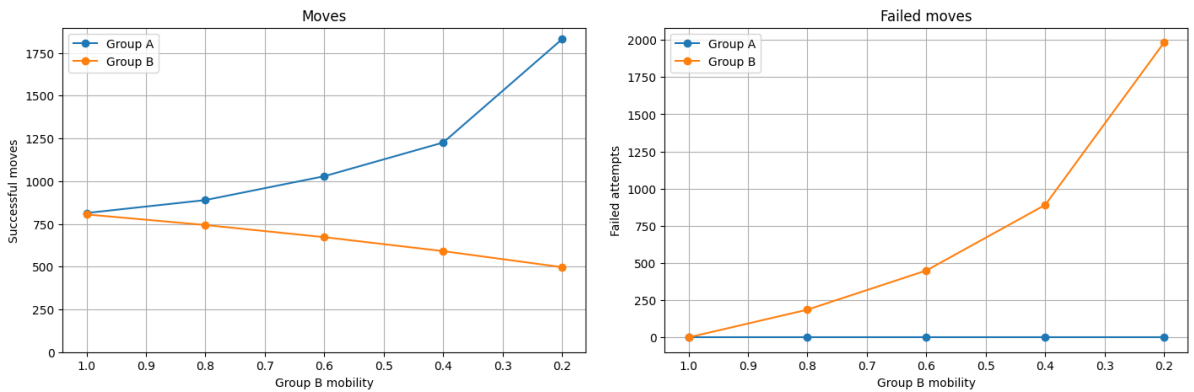


Figure 6: Successful(left) and failed(right) moves for different mobilities.

The mean successful moves of group A and group B move in opposite directions as group B mobility decreases. Group A's successful moves increase, while group B's successful moves decrease. Both start off at 805 moves, with group B quickly dropping to 496 moves, while group A increases to 1830 moves.

The failed moves we also demonstrates an increase in failed attempts; the lower group B mobility has a total of 1982 failed moves, while group A stays consistent at 0.

3.7 Tolerance thresholds through lower mobility(RQ5)

Figure 7 plots heatmaps, showing the steps and satisfaction gap for different tolerance thresholds and group B mobilities.

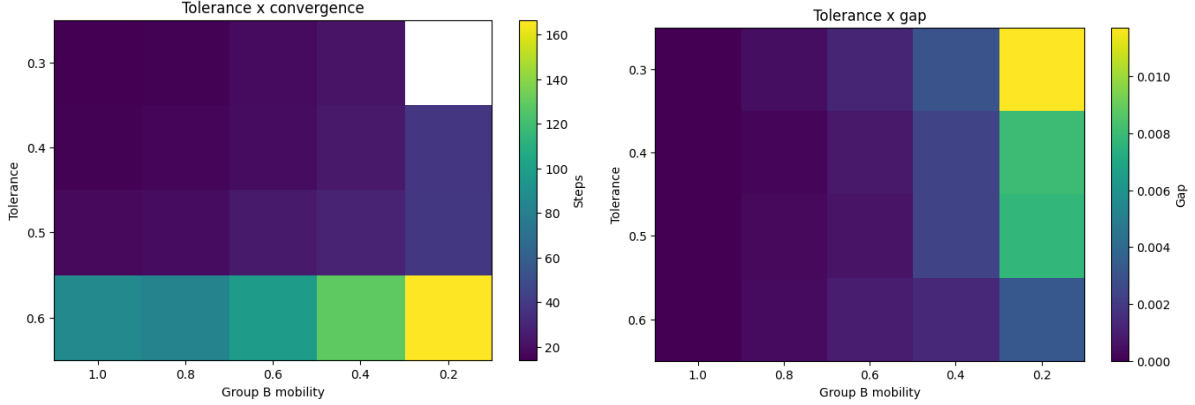


Figure 7: Convergence(left) and Gap(right) heatmaps for different mobilities and thresholds. White cells show no convergence.

Taking a look at the heatmaps, it is evident that having a lower tolerance and a higher mobility decreases the amount of steps needed for convergence. Additionally, for a tolerance of 0.3 and a mobility of 0.2, there is no convergence at all.

In contrast, the satisfaction gap shows that a lower tolerance and a lower mobility causes a larger satisfaction gap, while increasing the tolerance threshold, even with the same mobility, causes decreases the gap and therefore creates a more satisfied group B, meaning that RQ5 is only partially answered.

3.8 Distance restriction through lower mobility(RQ6)

The final subsection is answered by Figure 8. Here, segregation, convergence and satisfaction gap for a radius limited-group B contrasts to the previous group B, which could relocate anywhere on the grid. There, radius limitation was only 6 neighbouring cells.

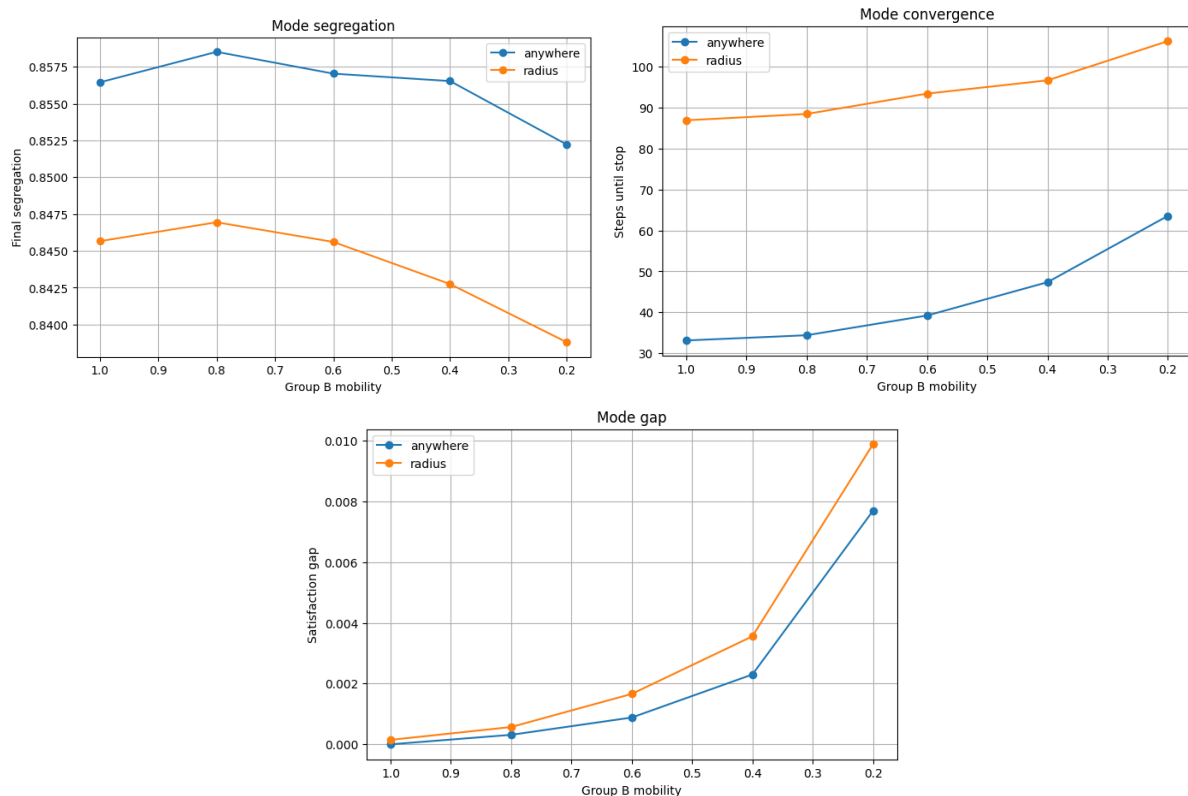


Figure 8: Segregation, Steps and Satisfaction for radius limited movement.

As can be seen in all three plots, radius limited movement changes the outcomes in every way. Segregation decreases, indicating a more mixed pattern overall. Convergence increases dramatically, increasing the number of steps to convergence by a large amount. The satisfaction gap also increases, showing that overall group B is less satisfied for the same mobilities.

4 Discussion

The results show that unequal mobility does not affect all segregation related outcomes in the same way. The clearest finding is that lower mobility for group B changes the process of reaching equilibrium more strongly than it changes the final segregation pattern.

Does lower mobility for one group increase the final residential segregation index?

As seen in the results, this question is clearly not supported in our testing, as all mobilities showed similar segregation for all repetitions. This suggests that in the model the schelling-type dynamics already lead to a high level of clustering, independent of equal mobility. Reducing the mobility for one group does not increase final segregation by a large margin. This does not make mobility constraints irrelevant, but suggests that their effect is not visible at the final segregation index.

Does lower mobility for one group increase the number of steps needed to reach equilibrium?

In contrast to RQ1, this is supported more clearly. Lower mobility for group B increases the amount of steps needed to reach equilibrium. This indicates that a lower mobility slows down the adjustment process. When a group is less able to move, the system needs more time to reach a stable state. This means that unequal mobility seems to affect the dynamics of segregation more than the final segregation itself.

Does lower mobility for one group increase the satisfaction gap between the high-mobility and low-mobility groups? This question is also supported, even if the effect is smaller than for convergence time or relocation attempts. The satisfaction gap clearly increases when the mobility of group B decreases. This means that group B with a lower mobility is on average less satisfied than a high mobility group A. Even given a similar final segregation, the process for the two groups does not happen equally. This means that the model shows a difference between global spatial outcomes and group level satisfaction outcomes.

Does lower mobility for one group increase failed relocation attempts for that group? This question is strongly supported. A lower mobility for group B clearly increases failed relocation attempts for that group, while at the same time, successful moves of group B decrease and successful moves of group A increase. This clearly shows that unequal mobility creates an unequal relocation constraint. Agents of group B are more often unable to move from their dissatisfaction, while group A agents move more often. This is one of the strongest indicators that the mobility constraint has a meaningful effect.

Does a higher tolerance threshold strengthen the effect of unequal mobility on convergence time and satisfaction inequality? This question is only partially supported. A higher tolerance threshold increases the effect of unequal mobility on convergence time, because the system needs more steps to reach equilibrium when agents require a higher share of same group neighbors. With regards to satisfaction, the effect is less consistent. Here, the gap is not just the largest at the highest tolerance threshold, which means that a higher tolerance does not automatically lead to an increase in every effect of unequal mobility. Instead, tolerance mainly increases the difficulty of reaching the equilibrium, while for satisfaction, it has the inverse effect of actually making it easier for agents to be satisfied with a lower tolerance.

Does restricted destination access, such as radius-limited movement, increase the effect of unequal mobility on convergence time and satisfaction inequality? This question is also supported for radius limited movement; restricting the destination access for group B increases convergence time and satisfaction gap compared with unrestricted movement. This also suggests that unequal mobility becomes even more consequential when it is not only modeled as a lower probability of moving, but also as a restriction on where agents can move. An interesting finding is that radius limited movement decreases the final segregation, which could be interpreted as a more mixed final pattern, but not strictly a better outcome for the constrained group, because satisfaction gap still increases. So lower segregation does not automatically mean that both groups are equally satisfied.

This model suggests that unequal mobility is most important for process level inequality, which could be missed by using just the final segregation index alone. While a lower mobility of a group does not significantly increase final segregation in the model, it does increase convergence time as well as failed relocation attempts and satisfaction inequality. The results show that residential segregation models should not only consider final patterns, but also unequal abilities to react to dissatisfaction during relocation.

5 Limitations

This project uses a simplified agent-based model, which has several limitations. The first limitation being that the model uses a two dimensional grid instead of real geographic data, which makes results easier to interpret but also means that real structures, like roads, housing density and public transport access are not correctly represented.

The second one is that the model only includes two groups of equal size. Usually residential segregation involves more than two groups with additional differences in income, household size, employment, etc. In this model both groups have the same tolerance thresholds, meaning that the only difference between them is mobility, which is useful for isolating the issue analyzed in this project, but also simplifies the overall concept.

As a third limitation, the model uses random relocation for all available cells. This means that the agents do not actively search for the best possible place. Future work could also include more detailed destination choices.

6 Conclusion

This project investigated whether unequal mobility between social groups affects the final segregation, convergence time, satisfaction gap and relocation constraints in a Schelling-type residential model. Results show that a lower mobility for group B does not meaningfully increase the final residential segregation, and final segregation stays high and relatively stable across the tested values.

Still, unequal mobility clearly has an effect on the process through which the system reaches stability. Lower group B mobility increases the number of steps, increases the satisfaction gap and increases failed relocation attempts. The results clearly show that group B is less able to act on the dissatisfaction, while group A performs as expected.

The tolerance analysis shows that a higher tolerance threshold increases the effect of unequal mobility on convergence time but not on satisfaction gap. Regarding satisfaction, the pattern is the inverse, where large gaps appear at lower tolerance levels when combined with lower group B mobility. This is further increased by the radius limited relocation.

The overall main conclusion is that unequal mobility matters more for process level inequality than for the final segregation alone. Looking only at that would miss important differences in convergence time, satisfaction and failed movement. This suggests that segregation models should not only be evaluated on segregation patterns but also on unequal response to dissatisfaction when relocating.

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